Mathuja Parameshwaran

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Professional Summary

Final-year Computer Science undergraduate with a strong foundation in software development, algorithms, and problem-solving. Experienced in building full-stack web applications and implementing efficient solutions using modern programming technologies. Skilled in Java, Python, and JavaScript with knowledge of software engineering principles, debugging, and system design. Seeking a Software Engineering Internship to contribute to developing scalable applications and gain industry experience.

Education

Bachelor of Science Honours in Computer Science (Reading), University of Jaffna — Nov 2022 – Present

- **Current OGPA: 3.46 / 4.0** for 73 out of 120 credits

GCE Advanced Level – Physical Science Stream, Uduppiddy American Mission College — 2018 – 2020

- Results: 3B

GCE Ordinary Level , J/T.V.M.V — 2007 – 2017

- Results: 9A

Research Experience

Deep Learning-Based EEG Emotion Detection — 2025 – Present
github.com/Mathuja991/EmotionDetection_EEG

- Developing a deep learning–based EEG emotion recognition system using the DEAP dataset.
- Implementing a neural network combining time-domain CNN features and frequency-domain bandpower features using PyTorch.
- Technologies: PyTorch, NumPy, Matplotlib, Deep Learning (CNN), Google Colab

Projects

Software System for Mathavam (Healthcare Platform) - Team Project — 2025
Live Demo (Vercel) *Role: Full Stack Developer*

- Developed a MERN-stack healthcare platform to manage autism patient assessments and medical records.
- Implemented authentication, role-based access control, and secure backend APIs.
- Built dashboards to monitor patient assessment progress.
- Technologies: React.js, Node.js, Express.js, MongoDB

Agent-Based Network File Management System - Team Project — 2025
github.com/Agent-Based-Centralised-File-Deletion *Role: Backend Developer*

- Developed a Python-based agent system to automate file management across laboratory computers.
- Implemented a master-client architecture where a master PC sends commands to delete programming exercise files on client PCs.
- Enabled centralized management of files for multiple programming languages including Python, Java, C, C++, and JavaScript.
- Technologies: Python, Network Programming, Client–Server Architecture

Hall Booking Management System - Individual Project 2025
Live Demo (Vercel) | github.com/Mathuja991/HotelBooking

- Developed a full-stack system for managing hall reservations and bookings.
- Implemented JWT authentication, booking conflict detection, and RESTful APIs.
- Built an admin dashboard for managing users and reservations.
- Technologies: React.js, Node.js, Express.js, MongoDB

Technical Skills

Programming: Java, Python, JavaScript
AI / Machine Learning: Deep Learning, Neural Networks, CNNs
Web Development: React.js, Node.js, Express.js, REST APIs, TailwindCSS
Databases: MongoDB, MySQL
Tools: Git, GitHub, MATLAB, VS Code, Colab, Docker, Cisco Packet Tracer

Soft Skills

Problem Solving, Team Collaboration, Communication, Time Management, Adaptability

Certifications

- Python for Beginners – University of Moratuwa (CODL) 2024
  Credential ID: qyDykJtRgf
- IBM – Introduction to Web Development with HTML5, CSS3, and JavaScript 2024
  Credential ID: d185a11eb18c47e7afef0d7129bf2868
- edX – Python Basics for Data Science 2023
  Credential ID: dc805808ecc14deaa68e3965395cfa6b
- Great Learning – Introduction to Cryptography 2023

Participation in National/International Hackathons

- **IEEE Xtreme** – 17.0 2023
- **CodeStorm 2.0** – Yarl IT Hub 2024

Attended Conferences & Workshops

- **NVIDIA Deep Learning Institute (DLI) workshop on Fundamentals of Accelerated Data Science** Dec 2025
  Department of Computer Science and Engineering, University of Moratuwa
- **YarlInsight 2.0 Summer School** Jul 2025
  IEEE Student Branch, University of Jaffna, University of Jaffna

  Focused on MERN Stack Development (MongoDB, Express.js, React.js, Node.js)
- **Sri Lanka Student Workshop on Computer Science (SL-SWCS'25)** Dec 2025
  Department of Computer Science, University of Jaffna

Languages

Tamil: Native / Bilingual Proficiency
English: Professional Working Proficiency

Referees

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